

	All data is for information only-----(Home)
	PRODUCT DATA of AISI 301
Material	AISI 301
Product Datasheet	301 Data Sheet
General Information	High work-hardening rate; applications requiring high strength and ductility. Type 301 is strengthened by cold working. If cold-worked Type 301 is subjected to temperatures above 480C, its room-temperature strength is reduced.
Precautions	Type 301 should not be used for extended periods at temperatures of 400 to 900C and should not be cooled slowly from higher temperatures through this range. Zinc and lead particles must be removed before heating to avoid embrittlement.
Effects of Space environment	The natural oxide layer is not repaired in vacuum when damaged. Cold welding might occur in contact to similar materials.
Chemical Composition	16-18Cr, 6-8Ni, <2Mn, <1Si, <0.15V, rem Fe
Name of Manufacturer	Many European suppliers.
	EXPERIENCE IN PRACTICE
Development Status	Commercial Product
Cost Range	Low
Lot Reproducibility	Good
Equivalent Designations	1.4310, X12CrNi77, 301S21, Z12CN17-08, X12CrNi17-07, BS 801 Grade A; AISI301
Space Experience	Good
	PHYSICAL PROPERTIES
Crystal Structure	Face centered cubic (fcc)
Specific Gravity	7.91
Phase changes	Heating or cold work may result in some transformation of austenite to ferrite (martensite) simultaneously with the precipitation of carbides
Magnetic Properties	Nonmagnetic
	ELASTIC PROPERTIES
Tensile Modulus	193 GPa
Shear Modulus	78 GPa
Poisson Ratio	0.27
	ELECTRICAL PROPERTIES

Electrical Volume Resistivity	7.15e-7 ohm.m
THERMAL PROPERTIES	
Thermal Conductivity	16.2 W/m.K
Thermal Expansion Coefficient	16.6 µstrain/°C
Specific Heat	0.456 J/g.K Remark: annealed specimens
Melting Range	1400 to 1410 °C
RELEVANT PROPERTIES FOR USE IN SPACE COMMUNITY	
General Corrosion Status	No specific corrosion protection required
Corrosion/Thermal	Intergranular attack in corrosive media is pronounced if this steel is sensitized by slow cooling through or by exposure at temperatures between 400 and 900 C. Annealing will eliminate this sensitized condition.
Atmospheric Corrosion	General corrosion resistance like that of all austenitic stainless steels.
<u>Stress Corrosion Status</u>	High Resistance to SCC: Table 1 Data Quality: Design Data
EMF vs SCE	-0.09 V Remark: Measured in 3.5% NaCl solution Data Quality: Design Data
<u>Bimetallic Corrosion</u>	Can be connected without any restriction to: Cr, Au, Pt, C, Rh, Ag, Gun-metal, P-bronzes, Sn-bronzes, Ti+alloys Can be connected in a non-controlled environment to the above and Cd, Cu, Brasses, Ni, monel, inconel, N-Mo alloys, 400 series stainless steel Can be connected in a clean room environment to above and Al-Cu alloys, Cu-Ni alloys, Al-bronzes, Si-bronzes Needs specific measures: All others Test:ECSS-Q-ST-70C Rev.1 Table 5.1 Data Quality: Design Data
Welding	This steel can be readily welded by all techniques, but the weld and heat affected zone becomes susceptible to intergranular corrosion, unless annealed after welding.
FLUID SYSTEMS COMPATIBILITY	
Gaseous Oxygen (GOX)	Meets the mechanical impact requirements. No batch test required.
Liquid Oxygen (LOX)	Meets the mechanical impact requirements. No batch test required.
Nitrogen Tetroxide (N2O4)	SCC susceptibility is low and no fluid decomposition

**Hydrazine
(HDZE)**

SCC susceptibility is low and no fluid decomposition